CYB 601: Software and Systems Security

Week – 0

Prerequisites and environment setup

by

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Get ready to build, break and defend one real system

- Goal: by Week 1, everyone can
- Reach the lab cloud
- Push to Github
- Run Kubectl
- Today:
  - Course Map
  - The Rules
  - Prerequisite refresher
  - Setup Lab

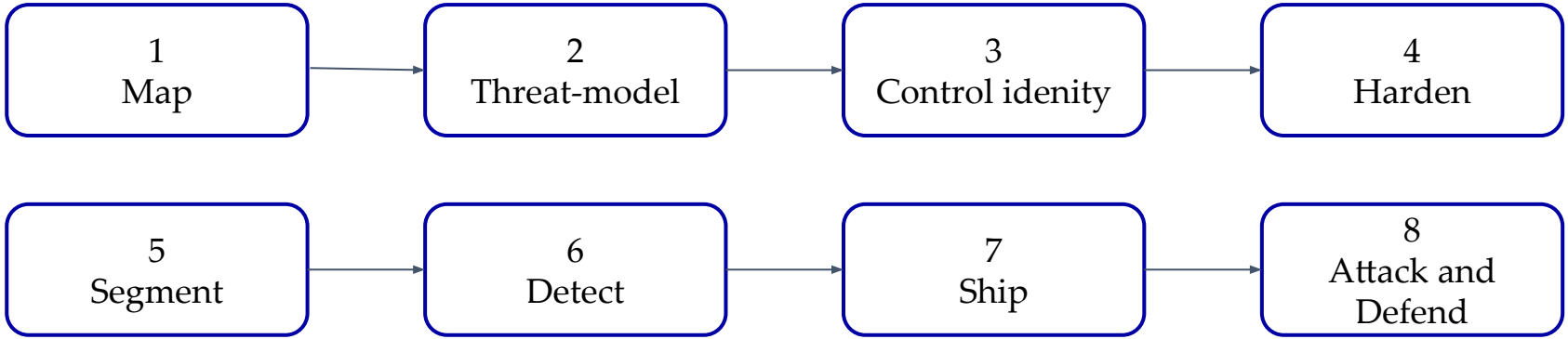






One system, eight weeks

- You are the security engineering team for a retailer taking a web shop to production.

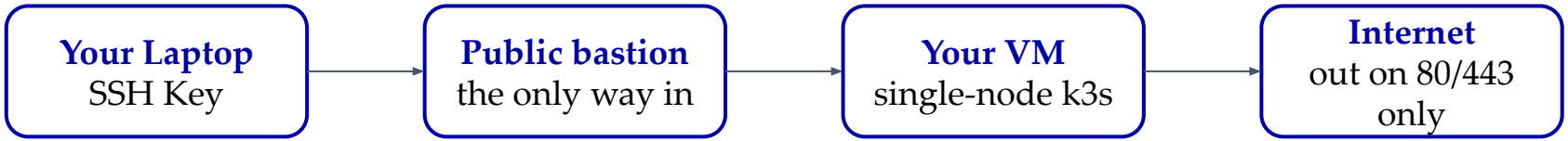


- Each lab's output feeds the next; the final plan cites all seven labs.

How you are graded

- University submission on GAP Portal, complete it, mostly Google Lab including Security topics.
- My Assignment on Github and Google Classroom.
- Labs are graded on evidence that rebuilds from your repository, not on screenshots.
- The final is a security plan plus a live defense to executives and engineers.

Your lab environment



- One Server (Proxmox):
  - 128 GB RAM
  - 32 Cores
  - 2 TB Storage
- Your VM:
  - 6 GB RAM
  - 2 vCores
  - 50 GB Storage
  - 1 Snapshot

Rules of engagement

- Attack only your own VM; in the capstone, only your assigned target
- Never scan or test anything outside the course cloud
- A secret that touches Git is leaked: revoke it first, then clean the history
- AI tools are fine for errors and syntax; the analysis is yours; disclose use in your README

What i assume you know

- Assumed:
  - Linux Command Linux
  - Git
  - Network Fundamentals
  - One Programming Language
- New for most, covered this week:
  - SSH jump hosts
  - kubectl
  - Terraform
- Readiness quiz: under 7 of 10 means booking extra time at the setup session

Linux you will use every week

- ssh, scp, sudo, systemctl, journalctl, ss, curl, grep, less
- Permissions decide who can read your private key, kubeconfig and secrets

```
-rw-------  ~/.ssh/id_ed25519        600  only you
-rw-r--r--  ~/.ssh/id_ed25519.pub    644  anyone can read
-rw-------  ~/.kube/config           600  only you
```

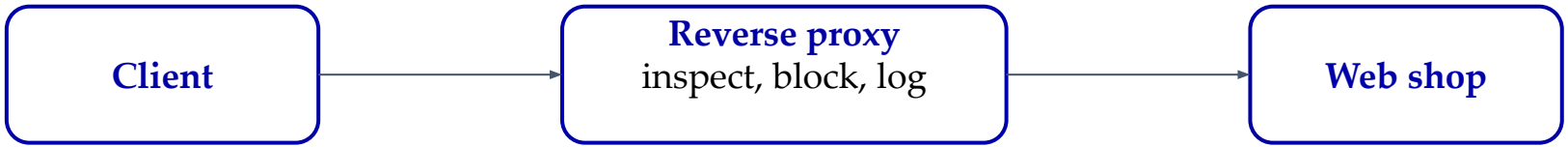
- Read logs before guessing

```
$ journalctl -u k3s -f
```

Networking refresher

- IP addresses, CIDR, ports, DNS; TCP and UDP
- TLS: what a certificate proves, and what it does not
- An HTTP request: method, path, headers, cookies, body
- A reverse proxy is one front door that can inspect, block and log

```
POST /cart HTTP/1.1          method, path
Host: shop.example Header
Authorization: Bearer eyJ…   Week 3 token
Cookie: session=4f2a…        cookie
{"sku":"TEE-42","qty":1}     body
```



Git the way this course uses it

- One private repository, a folder per lab

```
cyb601-lastname/
├── tools/week0-check.sh
├── lab0/
│   ├── check.txt
│   ├── quiz.md
│   └── notes.md
└── one folder per lab …
```

- Submit by tagging

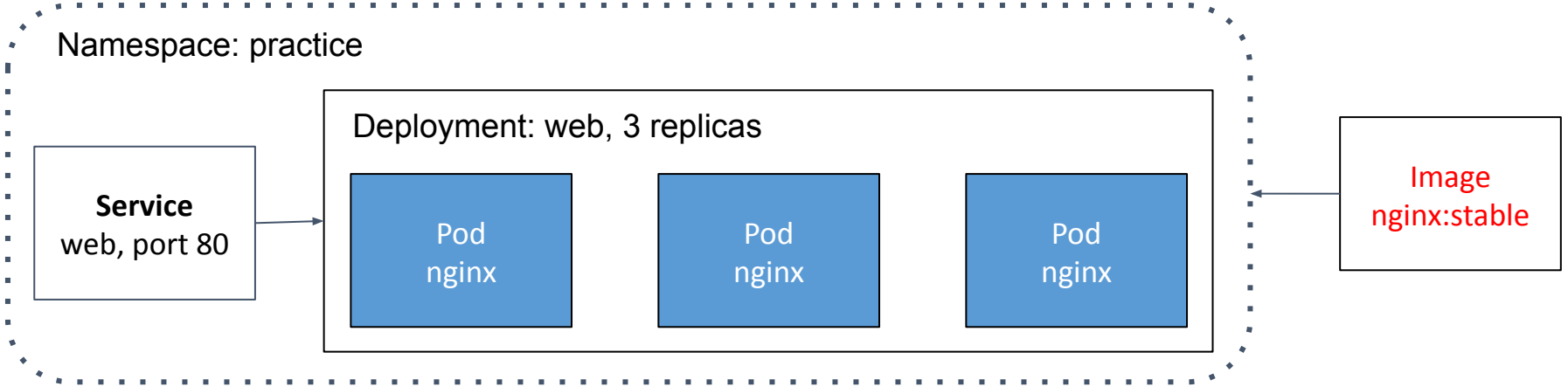
```
$ git tag lab1-submit
$ git push origin lab1-submit
```

- Never commit
  - API keys
  - Kubeconfigs
  - Terraform state

- Small commits with real messages; history is graded evidence.

Containers and Kubernetes

- Image, container, Pod, Deployment, Service, grouped in Namespaces



- kubectl verbs: get, describe, logs, exec, apply, delete
- K3s: A full Kubernetes in one small binary, sized for a 6 GB VM

Infrastructure as code

- Click-ops is: Unreviewed, Unrepeatable, Drifts silently, Big NO!
- Terraform or OpenTofu: plan, **Review**, apply, destroy

```
$ terraform plan
…
Plan: 1 to add, 0 to change, 0 to destroy.
```

- This week: You click through the Proxmox console.
- Week 1: You rebuild the same VM in code

Your Week 0 checklist

- Tools installed and versions printed (`kubectl version --client`)
- Bastion reachable over SSH, SSH key on GitHub (`ssh cyb601-bastion hostname`)
- Proxmox login, new password, API token stored safely(*Datacenter › Permissions › API Tokens*)
- Practice VM built, explored and expunged (`kubectl get nodes`)
- Repository created from the template, week0-submit tag pushed (`git push origin week0-submit`)

Getting help

- Ask with:
  - **The command + The full error + What you already tried**
- Bad help request

"SSH is broken. Can someone help?"

- Good help request:

"ssh cyb601-bastion hostname fails with Permission denied (publickey). ssh -v shows my key being offered, and I submitted the .pub yesterday. What should I check next?"

- Office hours and the forum; classmates may help debug, not do your work.

Thank you!
Any Questions?